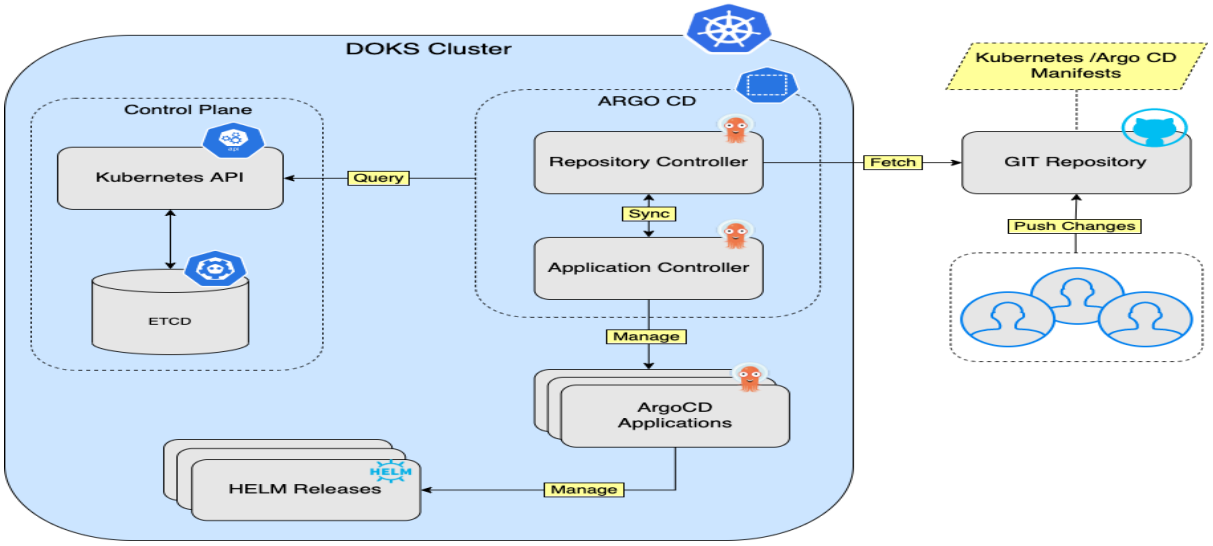
Argo CD



Overview:

- Collaborators (teams) create/update, and then push manifests to the designated Git repository.
- Argo CD notices the changes and performs cluster reconciliation.

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✅ 1. What is Argo CD?

- **Argo CD** stands for **Argo Continuous Delivery**.
- It's a **declarative GitOps tool** for Kubernetes.
- Argo CD continuously monitors your Git repositories and **automatically syncs** changes to Kubernetes clusters.
- It helps implement **GitOps** – using Git as the source of truth for infrastructure and application definitions.

✅ 2. Why Use Argo CD?

- Git-based single source of truth.
- Automates application deployment, rollback, and drift correction.
- Provides both **CLI** and **web UI**.
- Good visibility and auditability.
- Supports Helm, Kustomize, plain YAML, and Jsonnet.

✅ 3. Core Concepts

GitOps:

- Git is the source of truth for all deployments.
- Any change is done via Git commits.

Application:

- Main object in Argo CD.
- Defines what to deploy, where, and how.

Sync:

- Argo CD compares actual state (cluster) vs desired state (Git).
- If there's a difference, it syncs automatically or manually.

Health Status:

- Healthy, Degraded, Progressing, Missing.

Sync Status:

- Synced, OutOfSync, Unknown.

✅ 4. Argo CD Architecture

- **API Server** – Handles CLI/UI/API.
- **Repository Server** – Clones and reads Git repositories.
- **Application Controller** – Reconciles desired state with the live cluster.
- **Dex** (optional) – Provides authentication (SSO, LDAP, etc.).

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✅ 5. Installing Argo CD

Pre-requisites:

- Kubernetes cluster (e.g., Minikube, Kind, EKS).
- `kubectl` configured.

Install:

```
bash
kubectl create namespace argocd
kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
```

Access Argo CD UI:

```
bash
kubectl port-forward svc/argocd-server -n argocd 8080:443
```

Visit: `https://localhost:8080`

Get admin password:

```
bash
kubectl get secret argocd-initial-admin-secret -n argocd -o yaml
```

✅ 6. Creating Your First Application

Create a file `app.yaml`:

```
yaml
apiVersion: argoproj.io/v1alpha1
kind: Application
metadata:
  name: guestbook
  namespace: argocd
spec:
  project: default
  source:
    repoURL: https://github.com/argoproj/argocd-example-apps
    targetRevision: HEAD
    path: guestbook
  destination:
    server: https://kubernetes.default.svc
    namespace: default
  syncPolicy:
    automated:
      selfHeal: true
      prune: true
```

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Apply it:

```
bash
kubectl apply -f app.yaml
```

✅ 7. Argo CD CLI Basics

Login:

```
bash
argocd login localhost:8080
```

List apps:

```
bash
argocd app list
```

Sync app manually:

```
bash
argocd app sync guestbook
```

Delete app:

```
bash
argocd app delete guestbook
```

✅ 8. Sync Strategies

Manual Sync:
You manually click or run the command to sync changes.

Automatic Sync:

```
yaml
syncPolicy:
  automated:
    selfHeal: true
    prune: true
```

- **Self-heal**: Fix drift if someone changes resources manually.
- **Prune**: Deletes resources that were removed from Git.

✅ 9. Authentication and RBAC

- Argo CD supports **SSO** (GitHub, LDAP, Google, SAML, etc.) using **Dex**.
- You can configure roles and permissions using `argocd-rbac-cm` config map.

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✅ 10. Advanced Features

- **Helm support** – Use Helm charts as source.
- **Kustomize support** – Patch and overlay YAML files.
- **Multi-cluster support** – Deploy to different clusters.
- **App of Apps** pattern – Manage multiple Argo CD apps from a single Git repo.
- **PreSync / PostSync hooks** – Customize deployment process.

✅ 11. Example Use Case (Demo)

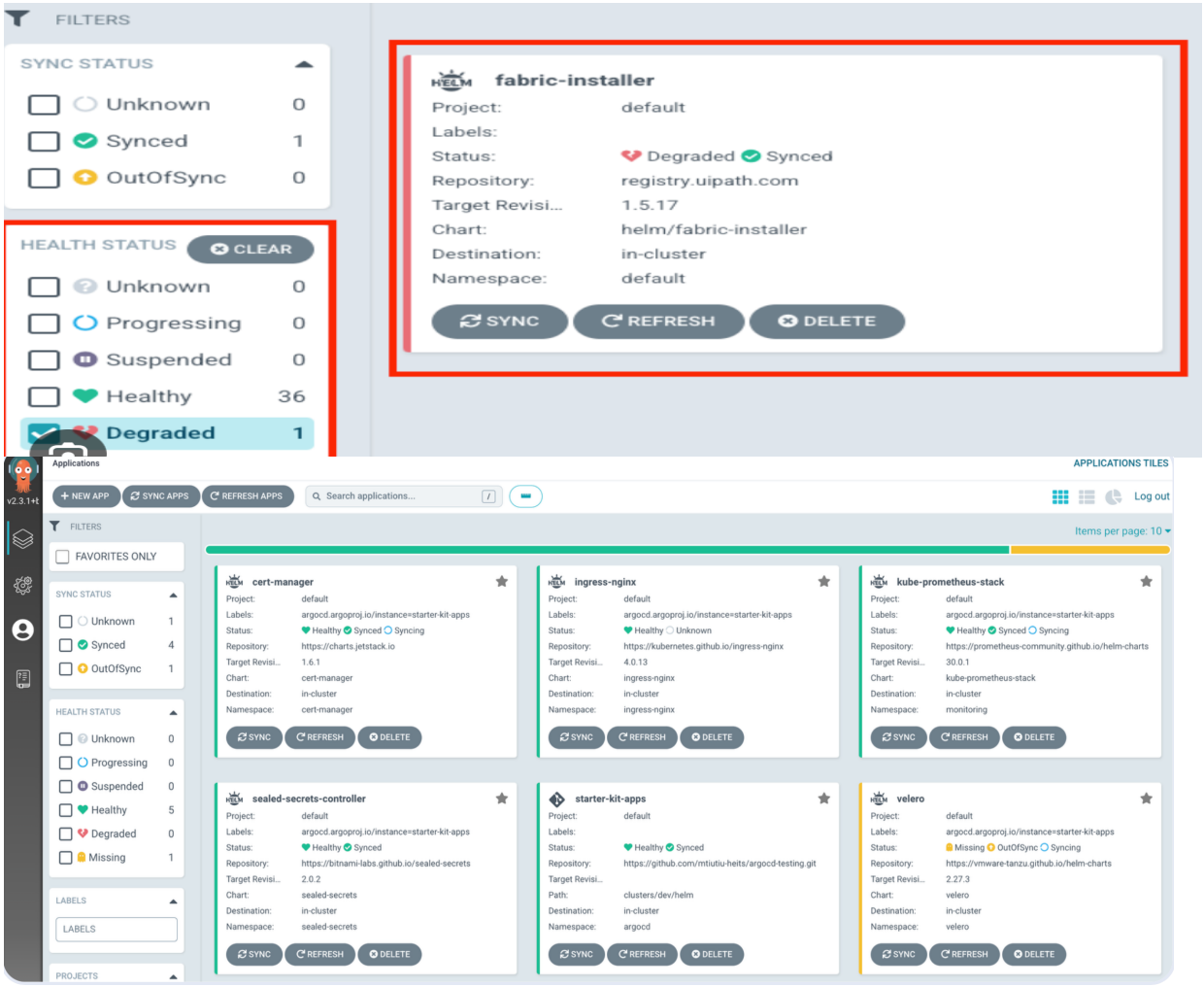
Steps:

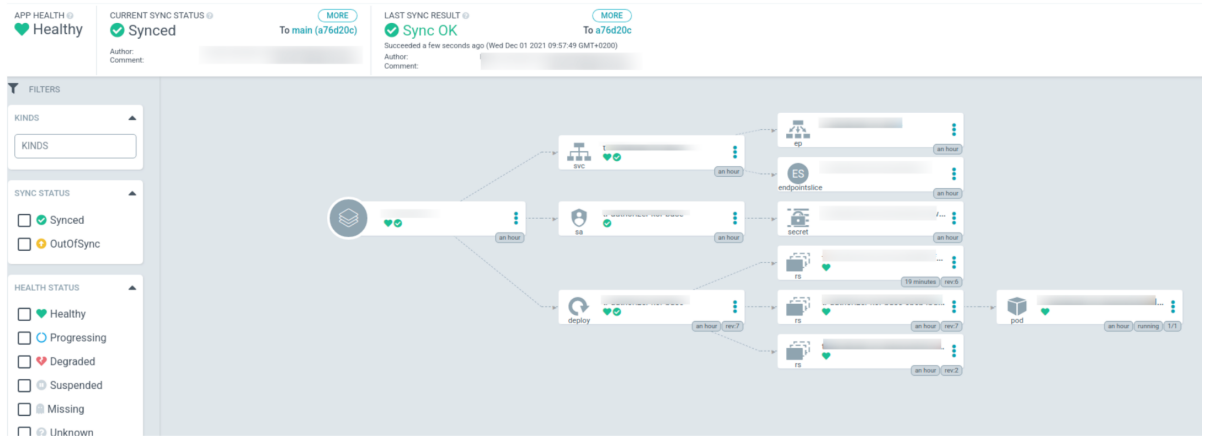
1. Deploy Guestbook app using the example repo.
2. Go to UI and watch the deployment.
3. Make a change in Git (e.g., change replica count).

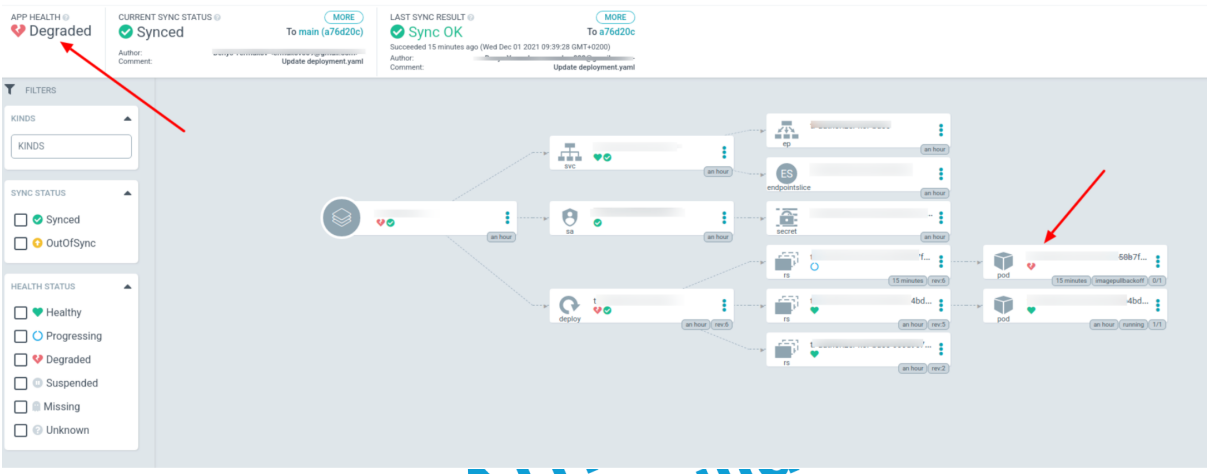
4. Commit and push the change.
5. Argo CD detects it and syncs automatically.
6. Manually delete a pod and watch Argo CD recreate it (self-heal).

✅ 12. Common Argo CD Commands

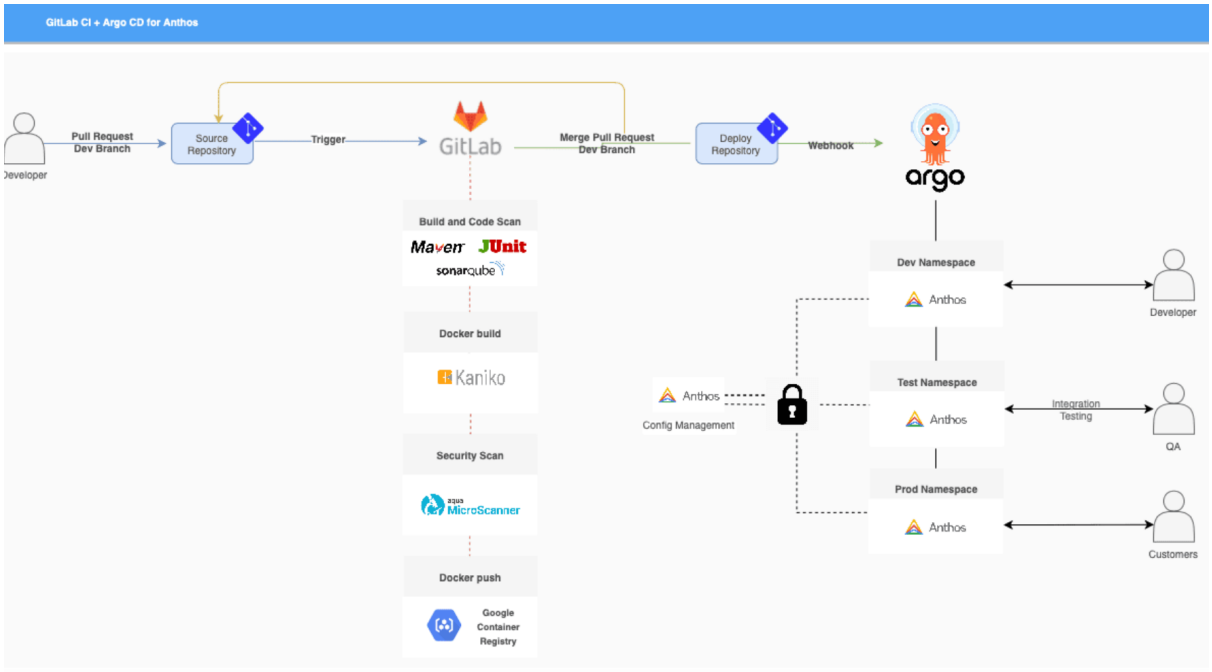
Command	Description
`kubectl get pods -n argocd`	Check Argo CD pods
`kubectl port-forward svc/argocd-server -n argocd 8080:443`	Access Argo CD UI
`argocd app create`	Create a new app
`argocd app sync <app-name>`	Trigger sync
`argocd app delete <app-name>`	Delete app







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